

Hao Zhou

School of Mathematical Sciences

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<https://hao23zh.github.io>

Research Interests

Numerical analysis; finite element methods; structure-preserving numerical methods; PDEs on manifolds.

Education

Peking University, Beijing, China

Ph.D. in Mathematics

2023 – Present

Thesis: *Finite element methods for fourth-order elliptic PDEs on surfaces* (proposal stage)

Adviser: Prof. [Shuonan Wu](#)

B.Sc. in Mathematics

2019 – 2023

Thesis: *Local discontinuous galerkin method for fractional Laplacian*

Adviser: Prof. Shuonan Wu

Publications

[Google Scholar Profile](#)

- [1] *A stabilized nonconforming finite element method for the surface biharmonic problem*. S. Wu, and H. Zhou; SIAM Journal on Numerical Analysis (2025). [DOI](#)
- [2] *A monotone discretization for the fractional obstacle problem and its improved policy iteration*. R. Han, S. Wu, and H. Zhou; Fractal and Fractional (2023). [DOI](#)

Preprints

- [1] *A Pressure-Free Virtual Element Method for the Surface Stokes Problem*, J. Hu, S. Xu, and H. Zhou; arXiv preprint (2026). [arXiv](#)
- [2] *A mixed finite element method for the Babuška paradox using only discrete geometry*, P. Tian, S. Wu, and H. Zhou; arXiv preprint (2026). [arXiv](#)
- [3] *Stabilized Morley FEM for surface Stokes in stream-function formulation: Optimal convergence via a new geometric estimate*, S. Wu, and H. Zhou; arXiv preprint (2026). [arXiv](#)
- [4] *Local discontinuous Galerkin method for the integral fractional Laplacian*, R. Han, S. Wu, and H. Zhou; arXiv preprint (2025). [arXiv](#)

Talks

- 1. CAST Young Scientists Leadership Program, Beijing, China, Aug. 2026.
- 2. Beijing–Tianjin–Hebei+ Conference on Computational Mathematics, Harbin, China, Aug. 2026.
- 3. CSIAM Young Forum, Chengdu, China, Jun. 2026.

Teaching Experience

Teaching Assistant, Linear Algebra I, Peking University

Fall 2026

Honors and Awards

- Outstanding Paper, Beijing–Tianjin–Hebei+ Conference on Computational Mathematics, 2026
- Young Science and Technology Talent Cultivation Project (Doctoral Program), China Association for Science and Technology (CAST), 2025
- Excellence Program in Applied Mathematics and Statistics, Peking University, 2023